信昌電子陶瓷股份有限公司
Prosperity Dielectrics Co., Ltd.
No.566-1, Gaoshi Rd., Lujhu, Yangmei Dist., Taoyuan City 32668, Taiwan, R.O.C.
Tel.：+886-3-4753355 Fax：+886-3-4854959

SPECIFICATION FOR APPROVAL

DATE：

CUSTOMER：

PART NAME： Metal Strip Type Halogen Free Current Sensing Resistors

CUSTOMER'S DWG. NO.：

CUSTOMER'S PART NO.：

PDC PART NO.： **FMF SERIES APPROVED**

DESCRIPTION.：

著作權所有 侵害必究
信昌電子陶瓷股份有限公司 Prosperity Dielectrics Co., Ltd.
PSA PASSIVE SYSTEM ALLIANCE
COPYRIGHT © PROSPERITY DIELECTRICS CO., LTD. ALL RIGHTS RESERVED.

ACTION / RESULT	" V "	CUSTOMER'S SIGNATURE	NOTE
FULL APPROVED			
CONDITIONAL APPROVED			
REJECTED			

OUR ACTION	SIGNATURE
PREPARED By	Jenny Tseng
CHECKED By	Tony Chou
APPROVED By	Byron Tsai

CUSTOMER SIGNATURE FOR ACCEPTANCE

Copyright © by Prosperity Dielectrics Co., LTD. | All rights reserved.

Contents.

Descriptions	Pages
01. Features	2
02. Applications	2
03. Dimension and construction	2
04. Power Derating Curve	2
05. Rating	3-4
06. Part Number	5
07. Marking / Soldering	5-6
08. Reliability Performance	7-9
09. Packing Information	9-10
10. Storage / Handling	11
※ Appendix (Max. Current &. Overload Current)	12-13

著作權所有 侵害必究
PSA PASSIVE SYSTEM ALLIANCE
Prosperity Dielectrics Co., Ltd.
COPYRIGHT © PROSPERITY DIELECTRICS CO., LTD. ALL RIGHTS RESERVED.

Copyright © by Prosperity Dielectrics Co., LTD. | All rights reserved.

FMF series (AEC-Q200)
Metal Strip High Power
Lead Free Current Sensing Resistors

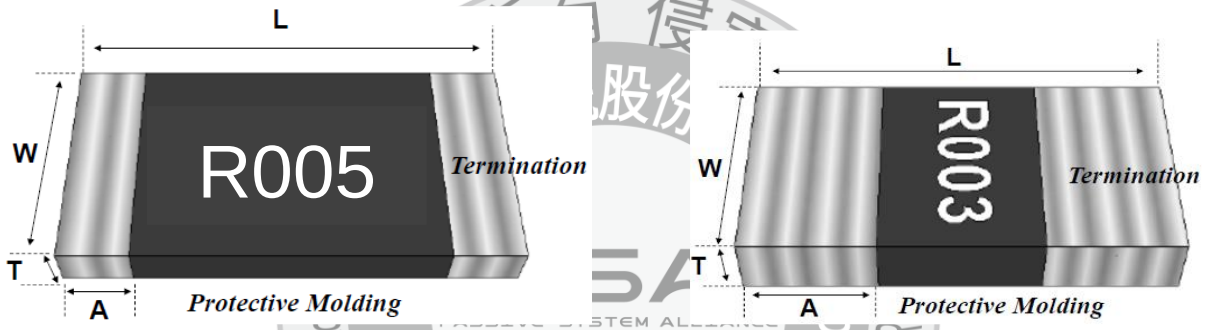
1. Features

- High power rating and low TCR.
- Low resistance and high precision (1%). Low EMF type ≦3uV/℃.
- Low inductance less than 1.0nH available please contact factory.
- Excellent reliability and suitable cost.
- Suitable for lead free soldering.
- RoHS compliant & Halogen Free.

2. Applications

- Switching model power supply.
- Battery pack.
- Notebook, Tablet PC
- Test Instrument.
- Power Amplifier.

3. Dimension and Construction



Item	Protective Molding	Resistive Element	Internal Terminal	External Terminal
Material	Resin	Alloy Metal	Copper	Solder

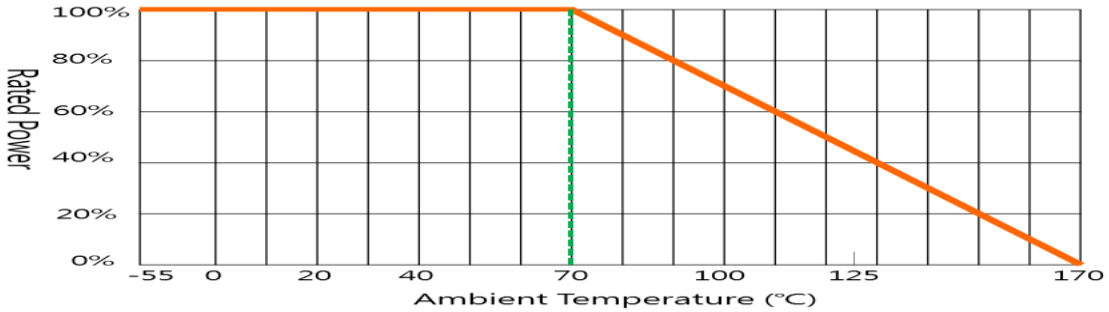
Unit：mm

FMF	L	W	T	A
1206 1~2mΩ	3.20±0.20	1.70±0.20	0.70±0.20	1.10±0.25
1206 0 &. 3~30mΩ	3.10±0.20	1.65±0.20	0.60±0.20	0.60±0.20
2512 0 &. 4~100mΩ	6.20±0.20	3.25±0.20	0.60±0.20	0.80±0.20
2512 1~3 mΩ	6.40±0.20	3.25±0.20	0.75±0.20	2.00±0.20
2512 3W 4~100mΩ	6.20±0.20	3.25±0.20	0.65±0.20	0.80±0.20
2512 3W 2~3 mΩ	6.40±0.20	3.25±0.20	0.75±0.20	2.00±0.20
2512 3W 1mΩ P/N. R001	6.40±0.20	3.25±0.20	0.80±0.20	2.00±0.20
2512 3W 0.5/0.75/1 mΩ P/N. R0L5/0L75/R1L0	6.30±0.20	3.25±0.20	0.80±0.20	2.25±0.20

Copyright © by Prosperity Dielectrics Co., LTD. | All rights reserved.

4. Power Derating Curve

Operating Temperature Range: -55 to +170℃



5. Rating

Table A.

Type \ Rating	Tolerance (%)	Rating 70℃ (W)	Max. Working Current (Voltage) ※1	Max. Overload Current (Voltage) ※1	Alloy Type	Temperature coefficient of Resistance (ppm/℃) ※2	Resistance (mΩ)
FMF06 1206	±1%(F) ±2%(G) ±5%(J)	0.5	22.4A (111mV)	50.0A (250mV)	Low EMF	±75	1,2
						±70	3,4,5,6,7,8,9, 10,12,15,20,25
			10.0A (111mV)	22.4A (250mV)	Standard	±50	5,10,15,18 20,25,30
		1	31.6A (158mV)	70.7A (354mV)	Low EMF	±75	1,2
						±70	3,4,5,6,7,8,9, 10,12,15,20,25
			14.1A (173mV)	31.6A (387mV)	Standard	±50	5,10,15 18,20,25,30
FMF25 2512	±0.5% (D) ※3 ±1%(F) ±2%(G) ±5%(J)	1	31.6A (158mV)	70.7A (354mV)	Low EMF	±70	1,2,2.5,3,4,5 10,15,20,25
			18.3A (469mV)	40.8A (1049mV)	Standard	±50	3,4,5,6,7,8,9,10 12,15,18,20,22,25 30,33,35,40,50,60 70,75,80,100
		2	44.7A (224mV)	100A (500mV)	Low EMF	±70	1,2,2.5,3,4,5 10,15,20,25
			25.8A (548mV)	57.7A (1225mV)	Standard	±50	3,4,5,6,7,8,9,10 12,15,18,20,22,25 30,33,35,40,50,60 70,75,80,100

Copyright © by Prosperity Dielectrics Co., LTD. | All rights reserved.

Table B.

Rating / Type	Tolerance (%)	Rating 70℃ (W)	Max. Working Current (Voltage) ※1	Max. Overload Current (Voltage) ※1	Alloy Type	Temperature coefficient of Resistance (ppm/℃) ※2	Resistance (mΩ)
FMF25 2512	**±0.5% (D)** ※3	**3**	77.5A **(47mV)**	173.2A **(106mV)**	Low EMF	**±70**	**0.5,0.75**
			54.8A **(245mV)**	122.5A **(548mV)**		**±70**	**1,2,2.5,3,4,5 6,7,8,9,10**
						±50	**20**
	±1%(F) ±2%(G) ±5%(J)	**3**	24.5A **(812mV)**	54.8A **(1817mV)**	Standard	**±70**	**5,6,8,<u>10</u>**
						±50	**12,14,15,16,18,20 25,30,33,35,40,50 60,75,80,100**

Note:

※1 : Related number are depend on specific items only. ※2 : TCR Hot (+25~+155℃).

※3 : ±0.5% available resistance with underline. Ex. **<u>10</u>**

1- E = $\sqrt{P*R}$ or Max. Working Voltage whichever is lower.

2- E：Working Voltage(V)，P：Rated Power (W)，R：Resistance Value(Ω)

3- Please keep the surface temperature do not exceed 105℃ when operating.

4- R-value might be variance depend on soldering conditions and please consider this influence before use milli-ohm resistors, and strongly suggest use the recommend solder pad to design your circuits.

5- Max. working &. Max. overload current details please refer Annex. 1.

T*able C.* Metal Jumper

Rating / Type & **PN**	Max. Working Current	Max. Overload Current	Resistance (mΩ)
FMF06 1206 **FMF06JTH000 -LH**	80A	100A	Max. **0.2**
FMF25 2512 **FMF25JPJ000 -LH**	120A	150A	Max. **0.1**

※Temperature Coefficient of Resistance is not applicable.

PDC

Copyright © by Prosperity Dielectrics Co., LTD. | All rights reserved.

6. Part Number

Type	Size	Tolerance	Packing	Watt	R Code	TCR	Special Code
FMF	**06** 1206 **25** 2512	**D** ±0.5% **F** ±1% **G** ±2% **J** ±5%	**T** Paper Tape 4Kpcs (1206) **P** Plastic Tape 4Kpcs (2512)	**F** 1/2W **H** 1W **J** 2W **K** 3W	**XXXX** 4 digits Jumper 000_ _ means blank	**-** As Rating table **X** Use for 2512≦3mΩ	General *Standard* **LH** *Low EMF* **BH** AEC-Q200 &. Anti-sulfur *Standard* **LHM** *Low EMF* **BHM**

※Anti-sulfur:△R≦1%. ASTM B-809,90'C,1000H &. EIA-977,105'C,750H Compliable.

Example：

FMF06FTHR010-BH

➔Metal strip, 1206 size, ±1%, paper tape, 1W, 10mΩ, low emf

FMF25FPKR002XBH

➔Metal strip, 2512 size, ±1%, plastic tape, 3W, 2mΩ, low emf, General

FMF25FPKR0L5XBH

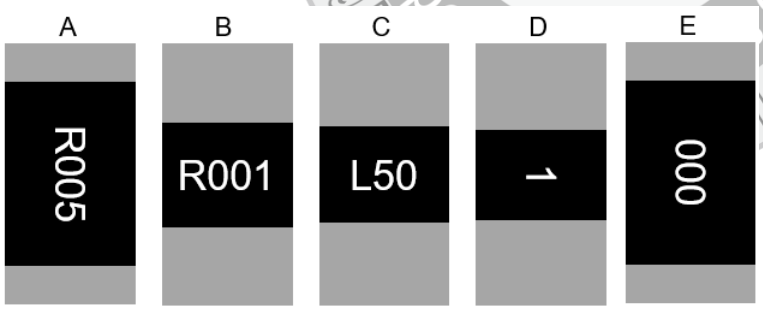
➔Metal strip, 2512 size, ±1%, plastic tape, 3W, 0.5mΩ, low emf, General

FMF25FPK0L75XBHM

➔Metal strip, 2512 size, ±1%, plastic tape, 3W, 0.75mΩ, low emf, AEC-Q200

7. Marking/Soldering Reference

Top side marking, marking format decided by P/N's R-Code.



A R□□□ 1206 R003~R030 / 2512 R004~R150

B R□□□ 2512 R001~R003

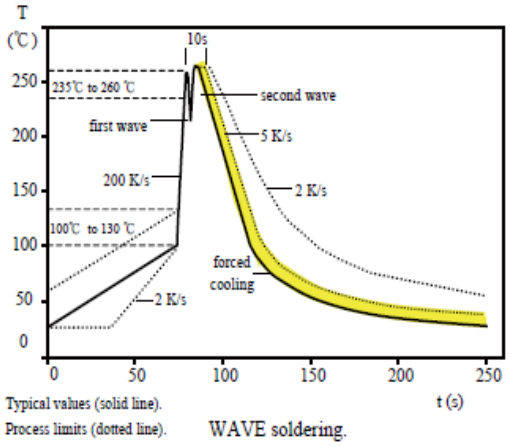
C □□□ 2512 R0L5(L50), 0L75(L75), R1L0(1L0)

D □ 1206 R001(1), R002(2)

E □□□ 1206 &. 2512, 000 (000)

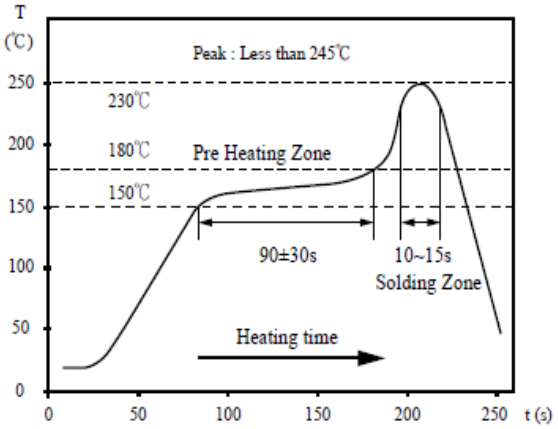
Copyright © by Prosperity Dielectrics Co., LTD. | All rights reserved.

Soldering Reference：Compatible for most industrial soldering request.



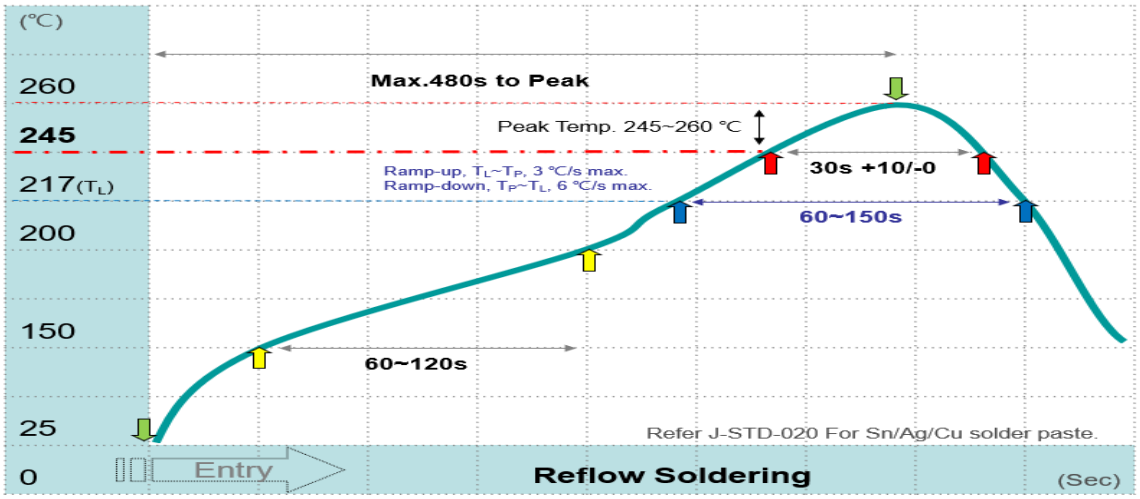
Typical values (solid line).
Process limits (dotted line).

WAVE soldering.



IR Reflow soldering.

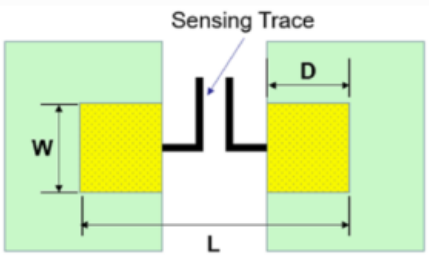
J-STD-020



Prosperity Dielectrics
PROSPERITY DIELECTRICS CO., LTD. ALL RIGHTS

Recommend Solder Pad Dimensions：unit.(mm)

Type	W	D	L
1206	1.80	1.30	4.70
1206 1~2 mR	1.80	2.30	5.60
2512	3.70	1.60	7.60
2512 0.5~3 mR	4.00	3.00	7.30



Copyright © by Prosperity Dielectrics Co., LTD. | All rights reserved.

8. Reliability Performance

8.1 AEC-Q200 type.

Test Item	Specification	Test Method (AEC-Q200. IEC 60115)
DC Resistance	J：±5%　G：±2% F：±1%　D：±0.5%	**AEC-Q200 TABLE 7.1** **IEC 60115-1 / JIS C 5201-1 , Clause 4.5** Measure the resistance Value.
High Temperature Exposure (Storage)	J、G：△R≦ ±3% F、D：△R≦ ±1%	**AEC-Q200 TABLE 7.3** 1000 hrs. @ T=170℃. Unpowered. Measurement at 24 ±2 hours after test conclusion.
Temperature Cycling	J、G：ΔR≦ ±1% F、D：ΔR≦ ±0.5% No mechanical damage.	**AEC-Q200 TABLE 7.4** 1000 Cycles (-55°C to +125°C). Measurement at 24±2 hours after test conclusion. **IEC 60115-1 Clause 4.19 for General Type** Repeat 5 cycles as follows -55℃(30min.)→25℃(2~3min.)→155℃(30min.)→25℃(2~3min.)
Moisture Resistance	J、G：△R≦ ±1% F、D：△R≦ ±0.5%	**AEC-Q200 TABLE 7.6** Test 65℃/80~100%RH/10Cycles. Measurement at 24±2 hours after test conclusion. (t=24hrs/cycle).
Biased Humidity	J、G：Δ R≦ ±3% F、D：Δ R≦ ±1%	**AEC-Q200 TABLE 7.7** 1000 hours 85℃/85%RH. 10% of operating power. Measurement at 24 ±2 hours after test conclusion.
Operational Life	J、G：Δ R≦ ±3% F、D：Δ R≦ ±1%	**AEC-Q200 TABLE 7.8** Test 1000hr @ T=125℃ at specified rated power. Measurement at 24±2 hours after test conclusion.
External Visual	No visual damage and refer PDC marking code.	**AEC-Q200 TABLE 7.9** Inspect device construction, marking and workmanship.
Physical Dimension	Within the spec.	**AEC-Q200 TABLE 7.10** Verify physical dimensions to the applicable device detail specification.

著作權所有 侵害必究 普電子陶瓷股份有限公司 Prosperity Dielectrics Co., Ltd. COPYRIGHT © PROSPERITY DIELECTRICS CO., LTD. ALL RIGHTS RESERVED

Copyright © by Prosperity Dielectrics Co., LTD. | All rights reserved.

Mechanical Shock	Within product specification tolerance and no visible damage.	**AEC-Q200 TABLE 7.13** Test Peak value:100g's,Wave:Hail-sine, Duration:6ms,Velocity:12.3ft/sec.
Vibration	No mechanical damage.	**AEC-Q200 TABLE 7.14** 5 g's for 20 min., 12 cycles each of 3 orientations. Test from 10-2000 Hz.
Resistance to Solder Heat	J、G：ΔR≦ ±1% F、D：ΔR≦ ±0.5% No mechanical damage.	**AEC-Q200 TABLE 7.15** Solder dipping @ 270℃±5℃ for 10sec.±1sec.
Thermal Shock	J、G：ΔR≦ ±1% F、D：ΔR≦ ±0.5% No mechanical damage.	**AEC-Q200 TABLE 7.16** -55 to 155℃/ dwell time 15min/ Max transfer time 20sec/ 300cycles.
ESD	ΔR≦ ±1% No mechanical damage.	**AEC-Q200-002** Test contact min. 1KV.
Solder Ability	Over 95% of termination must be covered with solder.	**AEC-Q200 TABLE 7.18** a)Baking 155℃ 4H, dipping 235℃ 5s b)Steam 1H, dipping 215℃ 5s c)Steam 1H, dipping 260℃ 7s
Flammability	Refer UL-94.	**AEC-Q200 TABLE 7.20** UL-94 V-0 or V-1 are acceptable
Board Flex	J、G：ΔR≦ ±1% F、D：ΔR≦ ±0.5% No mechanical damage.	**AEC-Q200 TABLE 7.21** Bending 2mm 2512.1206
Terminal Strength	No mechanical damage	**AEC-Q200 TABLE 7.22** Force 1 Kg for 60 seconds.
Anti-Sulfur	Δ R≦ ±1%	**ASTM-B-809-95(Modified)** Sulfur 1000 hours, 90±2°C **EIA-977(Test B)** Sulfur 750 hours, 105±2°C

Remarks 1. Jumper(0Ω) resistance specification according to Table C.

Copyright © by Prosperity Dielectrics Co., LTD. | All rights reserved.

8.2 Standard type.

Test Item	Specification	**Test Method (IEC 60115 / JIS C 5201-1)**
DC Resistance	J：±5%　G：±2% F：±1%　D：±0.5%	**IEC 60115-1 / JIS C 5201-1 , Clause 4.5** Measure the resistance Value.
Short Time Overload	J、G：ΔR≦ ±2% F、D：ΔR≦ ±1%	**IEC 60115-1 / JIS C 5201-1 , Clause 4.13** 5 × Rated power for 5 seconds Measure resistance after 30 minutes
Solderability	Over 95% of termination must be covered with Solder	**IEC 60115-1 / JIS C 5201-1 , Clause 4.17** After immersing flux, dip in the 235±2℃ molten solder bath for 3±0.5 sec.
Resistance to Solder Heat	ΔR≦±(1%+0.1mΩ) No mechanical damage	**IEC 60115-1/JIS C 5201-1 , Clause 4.18** With 260±5℃ for 10±1sec.
Temperature Cycle	J、G：ΔR≦ ±1% F、D：ΔR≦ ±0.5% No mechanical damage.	**IEC 60115-1/JIS C 5201-1 Clause 4.19** Repeat 5 cycles as follows -55℃(30min.)→25℃(2~3min.)→155℃(30min.)→25℃(2~3min.)
Load Life Humidity	J、G：Δ R≦ ±3% F、D：Δ R≦ ±1%	**IEC 60115-1 / JIS C 5201-1 , Clause 4.24** 40±2℃ with relative humidity 90% ~ 95% DC rated voltage for 1.5 hours On 30 minutes Off. Cycle repeated 1000 hours. (Not applicable if 3W R value<1mΩ)
Temperature Coefficient of Resistance (TCR)	Refer Rating Table.	**IEC 60115-1, Clause 4.8** Temperature：(T1. +25℃) ~ (T2. +155℃) TCR(ppm/℃) = $(R_2-R_1)/R_1\times1/(T_2-T_1)\times10^6$ (+25~ -55℃ please contact factory.)
Load Life	J、G：Δ R≦ ±3% F、D：Δ R≦ ±1%	**IEC 60115-1, Clause 4.25** Rated voltage for 1.5 hours then a pause 0.5 hours at T=70±2℃. Cycle repeated 1000 hours.
Insulation Resistance	Between termination and coating must over 1000MΩ	**IEC 60115-1, Clause 4.6** Test voltage：100±15V
Bending strength	J、G：ΔR≦ ±1% F、D：ΔR≦ ±0.5% No mechanical damage.	**IEC 60115-1 / JIS C 5201-1 , Clause 4.33** Resistance change after bended on the 90mm PCB. Bending :2mm

Remarks 1. Jumper(0Ω) resistance specification according to Table C.

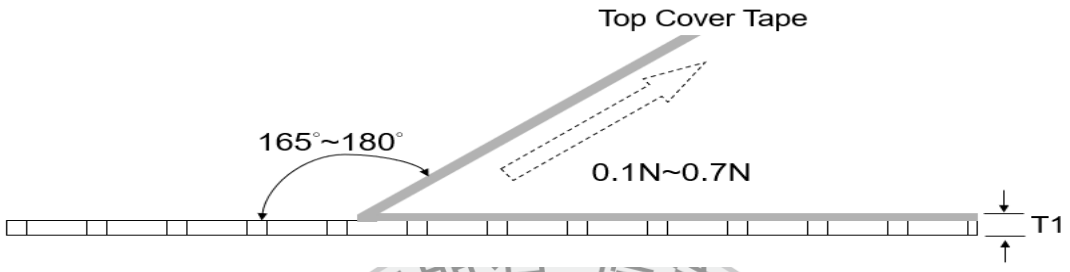
Copyright © by Prosperity Dielectrics Co., LTD. | All rights reserved.

9. PACKAGING

9.1 Peel Strength of Top Cover Tape

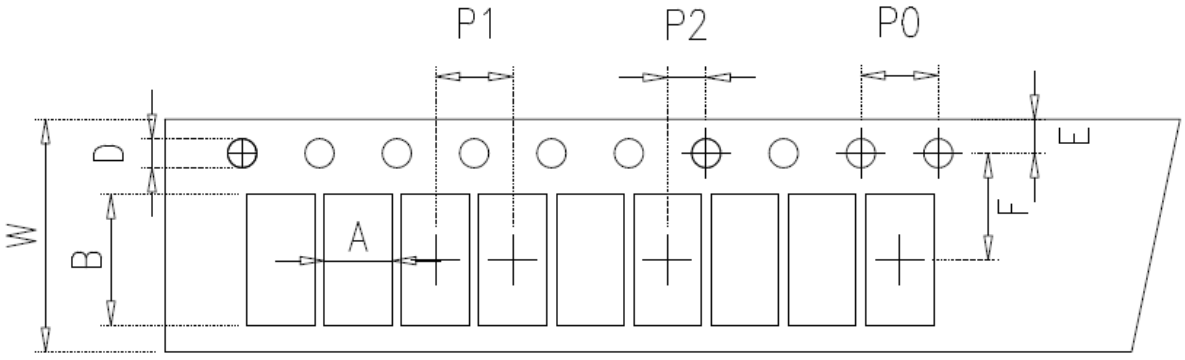
The peel speed shall be about 300 mm/min

The peel force of top cover tape shall between 0.1 to 0.7N



9.2 Tape Packaging Dimensions

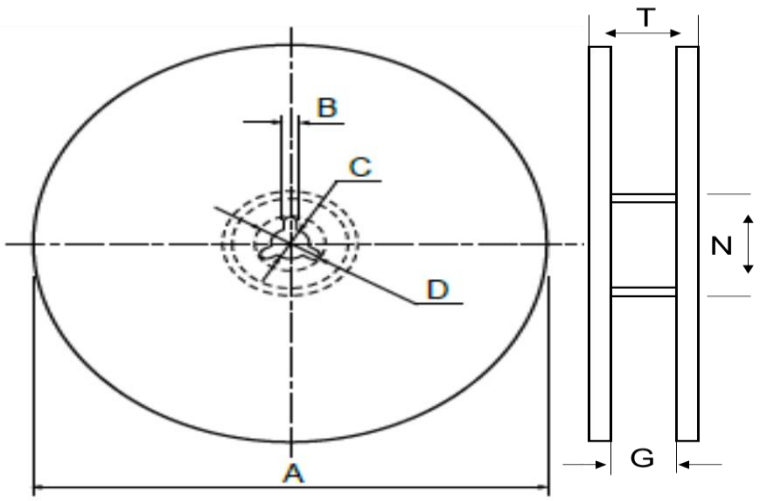
unit.mm



Size	A	B	W	F	E	P1	P2	P0	D	T1
1206	2.00±0.20	3.60±0.20	8.00±0.30	3.50±0.05	1.75±0.10	4.00±0.10	2.00±0.05	4.00±0.10	1.50±0.10	1.00±0.10
2512	3.50±0.20	6.75±0.20	12.0±0.30	5.50±0.05	1.75±0.10	4.00±0.10	2.00±0.05	4.00±0.10	1.50±0.10	1.15±0.10

Copyright © by Prosperity Dielectrics Co., LTD. | All rights reserved.

9.3 Reel Dimensions



Size	Packaging Q'ty	A	N	C	D	B	G	T
1206	4kpcs/Reel	178.0±2.0	60.0±0.5	13.0±0.5	20(Min.)	2.0±0.5	10.0±1.5	14.9max.
2512	4kpcs/Reel	178.0±2.0	60.0±0.5	13.0±0.5	20(Min.)	2.0±0.5	13.8±1.5	16.7max.

PSA
PASSIVE SYSTEM ALLIANCE
COPYRIGHT © PROSPERITY DIELECTRICS CO., LTD. ALL RIGHTS RESERVED

10. Storage &. Handling

… Products are recommended to be used up within one year as ensured shelf life.
Check solder ability in case shelf life extension is needed.

… To store products with following condition:
Temperature:5 to 40℃；Humidity: 20 to 70% relative humidity.

Precaution for use：

The AEC-Q200 series resistors is mainly used on general automotive equipment without safety considerations. Please contact our company in advanced if you intend to use resistor for designing the equipment which may damage itself and the safety of third party. If necessary, please consider to add the protect circuit in devising process and obtaining fully safety evaluation. The contents of the acknowledgment is only used for our parent company, marketing subsidiaries and official marketing agents who purchase our products. Not applicable for the other nonofficial channels.

※ All products specification and data are subject to change without notice.

Copyright © by Prosperity Dielectrics Co., LTD. | All rights reserved.

Annex.1 Max. working &. Max. overload current

2512 Rating Power 1.0W			2512 Rating Power 2.0W		
R_Value (mΩ)	Max. Working (A)	Max. Overload (A)	R_Value (mΩ)	Max. Working (A)	Max. Overload (A)
1	31.6	70.7	1	44.7	100.0
2	22.4	50.0	2	31.6	70.7
2.5	20.0	44.7	2.5	28.3	63.2
3	18.3	40.8	3	25.8	57.7
4	15.8	35.4	4	22.4	50.0
5	14.1	31.6	5	20.0	44.7
6	12.9	28.9	6	18.3	40.8
7	12.0	26.7	7	16.9	37.8
8	11.2	25.0	8	15.8	35.4
9	10.5	23.6	9	14.9	33.3
10	10.0	22.4	10	14.1	31.6
12	9.1	20.4	12	12.9	28.9
15	8.2	18.3	15	11.5	25.8
18	7.5	16.7	18	10.5	23.6
20	7.1	15.8	20	10.0	22.4
22	6.7	15.1	22	9.5	21.3
25	6.3	14.1	25	8.9	20.0
30	5.8	12.9	30	8.2	18.3
33	5.5	12.3	33	7.8	17.4
35	5.3	12.0	35	7.6	16.9
40	5.0	11.2	40	7.1	15.8
50	4.5	10.0	50	6.3	14.1
60	4.1	9.1	60	5.8	12.9
70	3.8	8.5	70	5.3	12.0
75	3.7	8.2	75	5.2	11.5
80	3.5	7.9	80	5.0	11.2
100	3.2	7.1	100	4.5	10.0

2512 Rating Power 3.0W		
R_Value (mΩ)	Max. Working (A)	Max. Overload (A)
0.5	77.5	173.2
0.75	63.2	141.4
1	54.8	122.5
2	38.7	86.6
2.5	34.6	77.5
3	31.6	70.7
4	27.4	61.2
5	24.5	54.8
6	22.4	50.0
7	20.7	46.3
8	19.4	43.3
9	18.3	40.8
10	17.3	38.7
12	15.8	35.4
14	14.6	32.7
15	14.1	31.6
16	13.7	30.6
18	12.9	28.9
20	12.2	27.4
25	11.0	24.5
30	10.0	22.4
33	9.5	21.3
35	9.3	20.7
40	8.7	19.4
50	7.7	17.3
60	7.1	15.8
75	6.3	14.1
80	6.1	13.7
100	5.5	12.2

Copyright © by Prosperity Dielectrics Co., LTD. | All rights reserved.

1206 Rating Power 0.5W			1206 Rating Power 1.0W		
R_Value (mΩ)	Max. Working (A)	Max. Overload (A)	R_Value (mΩ)	Max. Working (A)	Max. Overload (A)
1	22.4	50.0	1	31.6	70.7
2	15.8	35.4	2	22.4	50.0
3	12.9	28.9	3	18.3	40.8
4	11.2	25.0	4	15.8	35.4
5	10.0	22.4	5	14.1	31.6
6	9.1	20.4	6	12.9	28.9
7	8.5	18.9	7	12.0	26.7
8	7.9	17.7	8	11.2	25.0
9	7.5	16.7	9	10.5	23.6
10	7.1	15.8	10	10.0	22.4
12	6.5	14.4	12	9.1	20.4
15	5.8	12.9	15	8.2	18.3
18	5.3	11.8	18	7.5	16.7
20	5.0	11.2	20	7.1	15.8
25	4.5	10.0	25	6.3	14.1
30	4.1	9.1	30	5.8	12.9

著作權所有 侵害必究
台灣禾伸堂電子陶瓷股份有限公司 Prosperity Dielectrics Co., Ltd.
COPYRIGHT © PROSPERITY DIELECTRICS CO., LTD. ALL RIGHTS RESERVED.

Copyright © by Prosperity Dielectrics Co., LTD. | All rights reserved.